

Per-structure fidelity of CT-to-MR synthesis in deep gray matter

Tier 3 checkpoint `top_model_epoch_472.pth`

Abstract

Whole-brain image metrics are insensitive to the subcortical structures that matter for deep brain stimulation targeting: the thalamus is roughly 7 cm^3 of a 1200 cm^3 brain, so a model can place the deep gray badly and barely move the headline figure. We report three per-structure measurements on the Tier 3 CT-to-MR checkpoint over 36 held-out subjects: the change in segmented volume, the Dice overlap against ground truth, and the displacement of each structure’s centre of mass. Structures are placed accurately (mean centroid error 1.1 mm, approximately one voxel) but are systematically under-filled (mean volume ratio 0.911; 345 of 432 structure-subject pairs under-segmented). The volume deficit scales inversely with boundary contrast, being largest for the pallidum (-16.4%) and absent for the thalamus ($+0.3\%$). Together with a complex-wavelet SSIM gap of zero, this localises the residual error to intensity and texture within correctly placed structures rather than to misregistration.

1 Method

Both the real and the synthetic volume of each subject are segmented with SynthSeg, using the same per-subject mode recorded in the Phase 1 quality-control manifest, so that a difference between the two reflects the synthesis rather than the processing chain. Six bilateral structures are scored, left and right separately, giving 12 labels per subject and $36 \times 12 = 432$ structure-subject pairs. A structure that SynthSeg cannot find in the synthetic volume is recorded as Dice 0 rather than dropped; no such failures occurred in this cohort. Voxels are approximately 0.75 mm^3 .

Each structure is additionally exported as its own binary mask, named

`<pid>_<structure>_real.nii.gz`, `<pid>_<structure>_syn.nii.gz`,

carrying the segmentation affine so that the masks remain registered to the exported volumes.

Standard errors below treat the *subject* as the independent unit ($n = 36$), not the structure instance ($n = 72$), because the left and right instances of a structure within one brain are correlated.

2 Volume change

Five of the six structures lose volume. Across all 432 pairs, 345 (79.9%) are under-segmented, against a chance expectation of 216, so the bias is directional rather than scatter. The cohort median ratio is 0.921.

3 Dice overlap

Dice is not saturated. The noise floor is tight (SD 0.007–0.015) relative to the spread of the synthetic scores (SD 0.022–0.087), so the harness resolves a change of about 0.01 Dice at $n = 36$. The pallidum is the weakest structure by a wide margin: 0.731 ± 0.015 against the thalamus at 0.909 ± 0.004 , a gap of roughly 12 standard errors.

Table 1: Segmented volume of each structure in the real and the synthetic volume. Counts are means over 72 instances (36 subjects, left and right). The ratio column is the mean of per-instance ratios, which is the comparable quantity across structures that differ by an order of magnitude in size.

Structure	Voxels		Δ	$\Delta \%$	Volume (mm ³)		Ratio
	Real	Synth.			Real	Synth.	
Thalamus	9064	9090	26	0.3	6805	6808	0.999 ± 0.065
Caudate	5054	4778	-276	-5.5	3808	3600	0.948 ± 0.078
Putamen	6715	5826	-889	-13.2	5037	4374	0.874 ± 0.101
Pallidum	2092	1748	-344	-16.4	1572	1308	0.839 ± 0.143
Hippocampus	5640	5096	-545	-9.7	4231	3826	0.908 ± 0.075
Amygdala	2358	2110	-248	-10.5	1768	1584	0.899 ± 0.112
Cohort	—	—	—	—	—	—	0.911

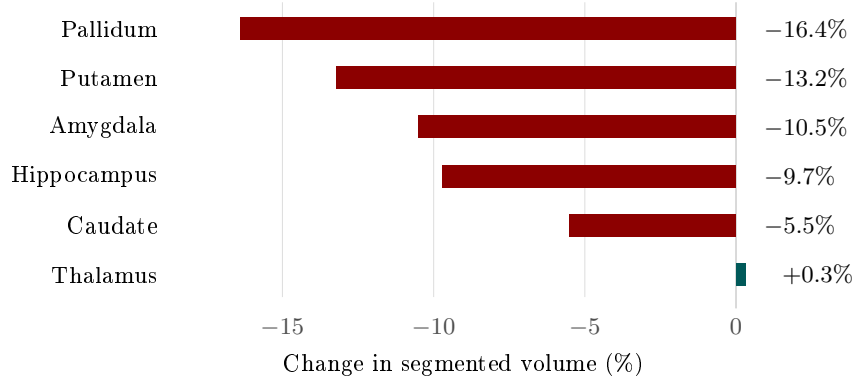


Figure 1: Volume change per structure. Bars grow leftward from zero because almost every structure shrinks. The thalamus, the structure with the most distinct borders, is the lone exception; the pallidum and putamen, which share a low-contrast boundary with each other, lose the most.

Table 2: Dice coefficient between the structure segmented from the real volume and from the synthetic volume. The noise floor is the Dice obtained by re-segmenting the *real* MR against the Phase 1 transported labels: identical anatomy, no synthesis. Headroom is the difference, i.e. the overlap still recoverable by improving synthesis.

Structure	Dice	Noise floor	Headroom
Thalamus	0.909 ± 0.022	0.977	0.068
Caudate	0.865 ± 0.046	0.971	0.105
Putamen	0.831 ± 0.046	0.967	0.136
Pallidum	0.731 ± 0.087	0.959	0.228
Hippocampus	0.818 ± 0.046	0.966	0.148
Amygdala	0.835 ± 0.051	0.959	0.125
Mean	0.831	0.966	0.135

4 Centroid displacement

Table 3: Displacement of each structure’s centre of mass between the real and the synthetic segmentation, in millimetres, computed with true voxel spacing. Image metrics are restricted to the structure: SSIM and CW-SSIM over its padded bounding box, PSNR masked to the structure itself.

Structure	Centroid (mm)	SSIM	CW-SSIM	PSNR (dB)
Thalamus	0.87 ± 0.34	0.680	0.658	26.12
Caudate	1.00 ± 0.63	0.728	0.689	24.01
Putamen	1.08 ± 0.47	0.646	0.652	26.21
Pallidum	1.50 ± 0.61	0.655	0.662	28.38
Hippocampus	1.23 ± 0.67	0.583	0.608	20.95
Amygdala	0.98 ± 0.45	0.554	0.576	23.74
Mean	1.11	0.641	0.641	24.90

Every structure lies within approximately one voxel of ground truth.

5 Interpretation

CW-SSIM is constructed to be tolerant of small translations: a shift rotates the phase of a complex wavelet coefficient while leaving its magnitude essentially unchanged, and the index scores phase *consistency* across a neighbourhood rather than absolute phase. The implementation is verified translation-tolerant (0.765 versus 0.636 for plain SSIM under a 2-pixel shift). Over the scored structures, CW-SSIM and SSIM are nevertheless identical (0.641 versus 0.641, Table 3): there is no misalignment for CW-SSIM to forgive. The residual error is therefore intensity and texture *within* the structure, consistent with the sub-voxel centroid displacements.

This resolves an otherwise anomalous result. The pallidum has the worst Dice and the worst centroid displacement, yet the *highest* PSNR of any structure at 28.38 dB. The pallidum is delimited by a low-contrast boundary against the putamen rather than by a strong edge. Mild compression of that contrast moves where the segmentation places the boundary, costing Dice and volume, while leaving pixel-wise error small. The putamen, its neighbour across the same weak border, is the second most affected structure; the thalamus, with the most distinct borders, shows no volume deficit at all. The ordering of the volume deficit by boundary contrast (Figure 1) is the direct evidence for this mechanism. Clinically this is the least convenient failure mode available, as the globus pallidus internus is an established deep brain stimulation target.

6 Limitations

The measurements are of a segmenter, not of anatomy. Every figure compares SynthSeg applied to the real volume against SynthSeg applied to the synthetic one. The noise floor bounds the segmenter’s own contribution at 0.966 ± 0.01 Dice, so approximately 3–4 points of the reported deficit are attributable to the tool rather than to the model. It follows that “the model under-fills the pallidum by 16%” and “SynthSeg is conservative on synthetic contrast” are not separable from these measurements alone; the defensible statement is that synthetic volumes *segment* smaller.

The synthetic volumes are generated under a favourable protocol. The model is conditioned on a global intensity histogram of the target MR, and in this evaluation that histogram is the ground-truth target’s own. This is defensible if framed as operator-supplied scanner calibration, but it is not a deployment condition, and it inflates every figure downstream. A re-evaluation conditioned on the training-set mean histogram, or on a template, yields the deployment-realistic figure and should be reported alongside.

Exclusions. Ten of the 2160 structure-subject pairs in the full 180-subject cohort (0.46%) are excluded by the frozen Phase 1 quality-control manifest, where the reference segmentation engulfs a lesion or abnormal tissue. No subject is dropped and the test split remains $n = 36$; all 432 pairs in this evaluation were scored successfully.